Angular Demos

ng new, ng serve,
(ng-)bootstrap library
basic elements: button, menu, tab, ..

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Course Esilv 2023

This document:
https://github.com/Arnaud-Nauwynck/presentations/web/
angular-demos-1-ng-new-ngbootstrap.pdf

2 pre-requisites:

1/ install **node.js**
2/ **$ npm install -g @angular/cli**

Also recommended :
3/ have 1G disk free on your PC
4/ install Intellij or Visual Studio Code
5/ network bandwidth
6/ patience

$ ng new

Create a new application

Upgrade existing @angular/cli .. If needed

```
$ ng update @angular/cli
```

or

```
$ npm uninstall @angular/cli
$ npm install –g @angular/cli
```

Upgrade existing angular project … if needed

$ ng update

```
$ ng update
Using package manager: npm
Collecting installed dependencies...
Found 22 dependencies.
    We analyzed your package.json and everything seems to be in order. Good work!
```

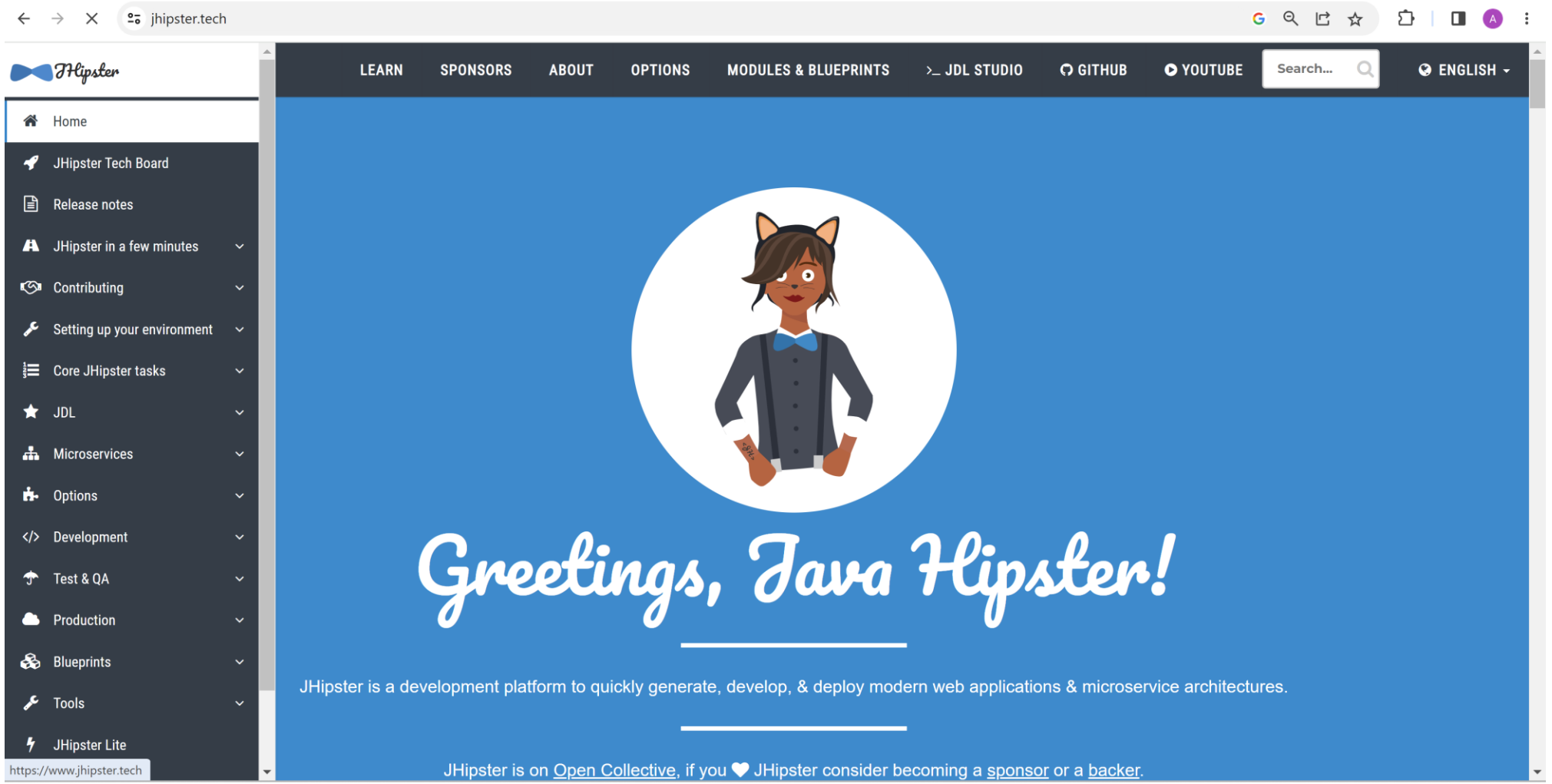
« ng new » alternatives

1/ $ **git clone** https://github.com/${org}/${repo}

2/ **jhispter**

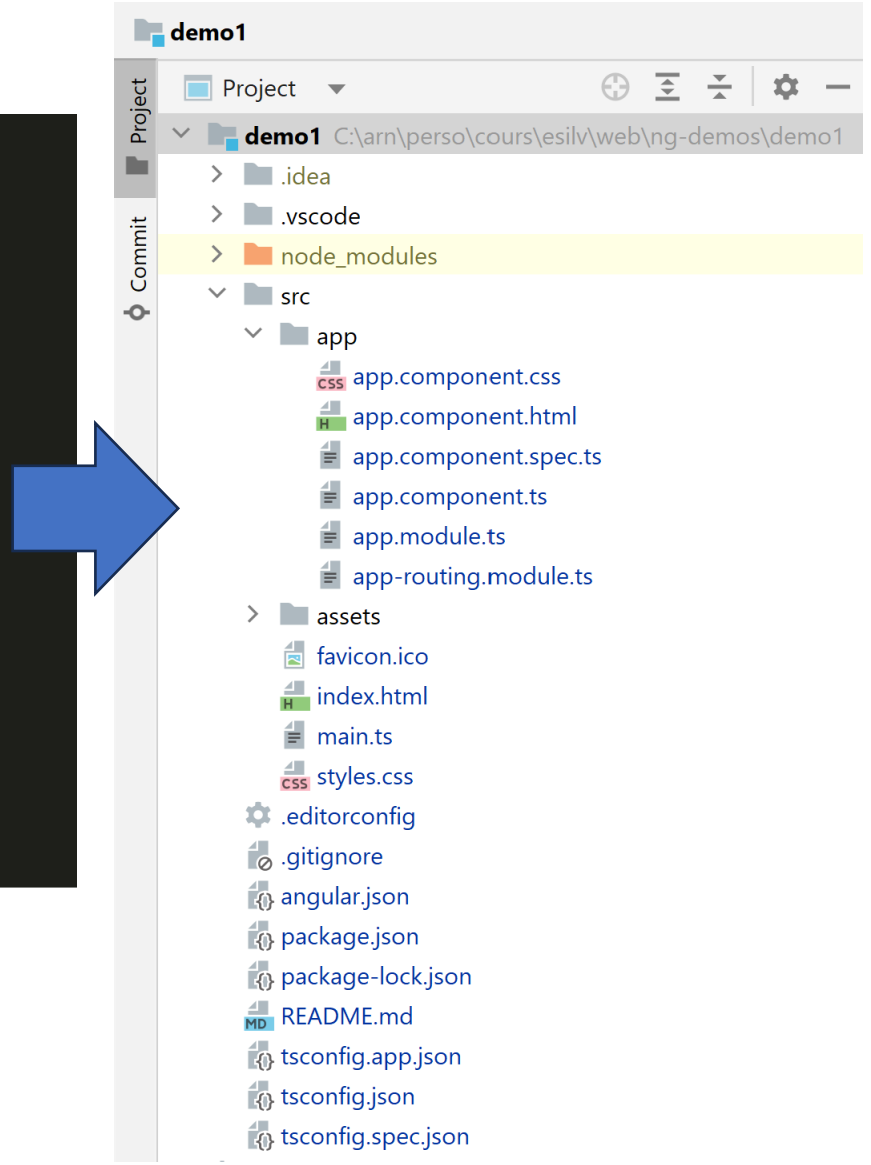
3/ read docs, edit package.json + angular.json
… NOT recommended

Jhipster = Java springboot(backend) + angular(frontend) app generator



ng new

```
$ ng new
? What name would you like to use for the new workspace and initial project? demo1
? Would you like to add Angular routing? Yes
? Which stylesheet format would you like to use? CSS
CREATE demo1/angular.json (2695 bytes)
CREATE demo1/package.json (1036 bytes)
CREATE demo1/README.md (1059 bytes)
CREATE demo1/tsconfig.json (901 bytes)
CREATE demo1/.editorconfig (274 bytes)
CREATE demo1/.gitignore (548 bytes)
CREATE demo1/tsconfig.app.json (263 bytes)
CREATE demo1/tsconfig.spec.json (273 bytes)
CREATE demo1/.vscode/extensions.json (130 bytes)
CREATE demo1/.vscode/launch.json (474 bytes)
CREATE demo1/.vscode/tasks.json (938 bytes)
CREATE demo1/src/favicon.ico (948 bytes)
CREATE demo1/src/index.html (291 bytes)
CREATE demo1/src/main.ts (214 bytes)
CREATE demo1/src/styles.css (80 bytes)
```



$ ng serve => http://localhost:4200

```
$ ng serve
√ Browser application bundle generation complete.

Initial Chunk Files   | Names         |  Raw Size
vendor.js             | vendor        |   2.04 MB |
polyfills.js          | polyfills     | 314.82 kB |
styles.css, styles.js | styles        | 209.94 kB |
main.js               | main          |  48.08 kB |
runtime.js            | runtime       |   6.51 kB |

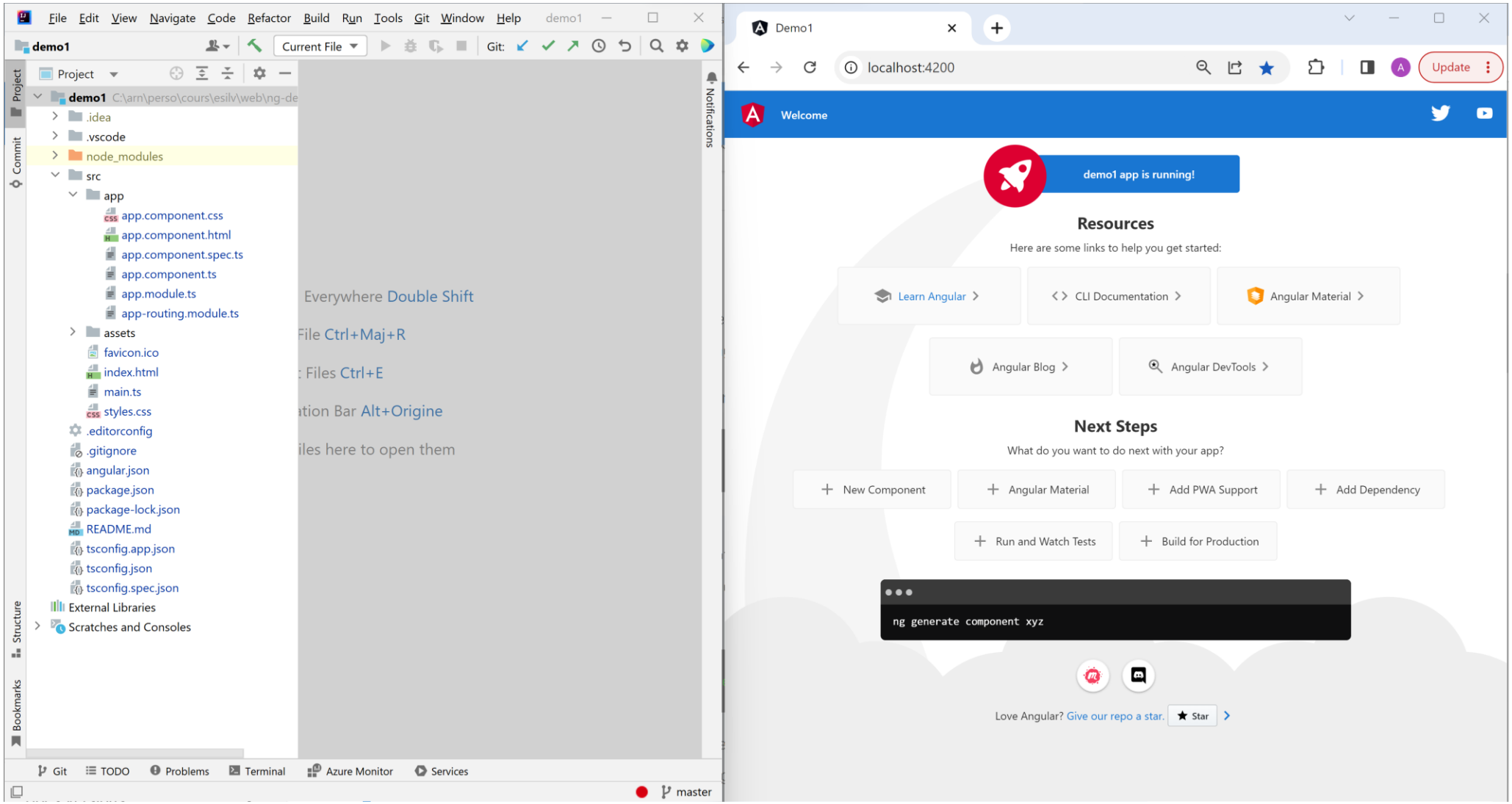
                      | Initial Total |   2.61 MB

Build at: 2023-07-09T10:06:15.879Z - Hash: 70e19beef187b407 - Time: 22482ms

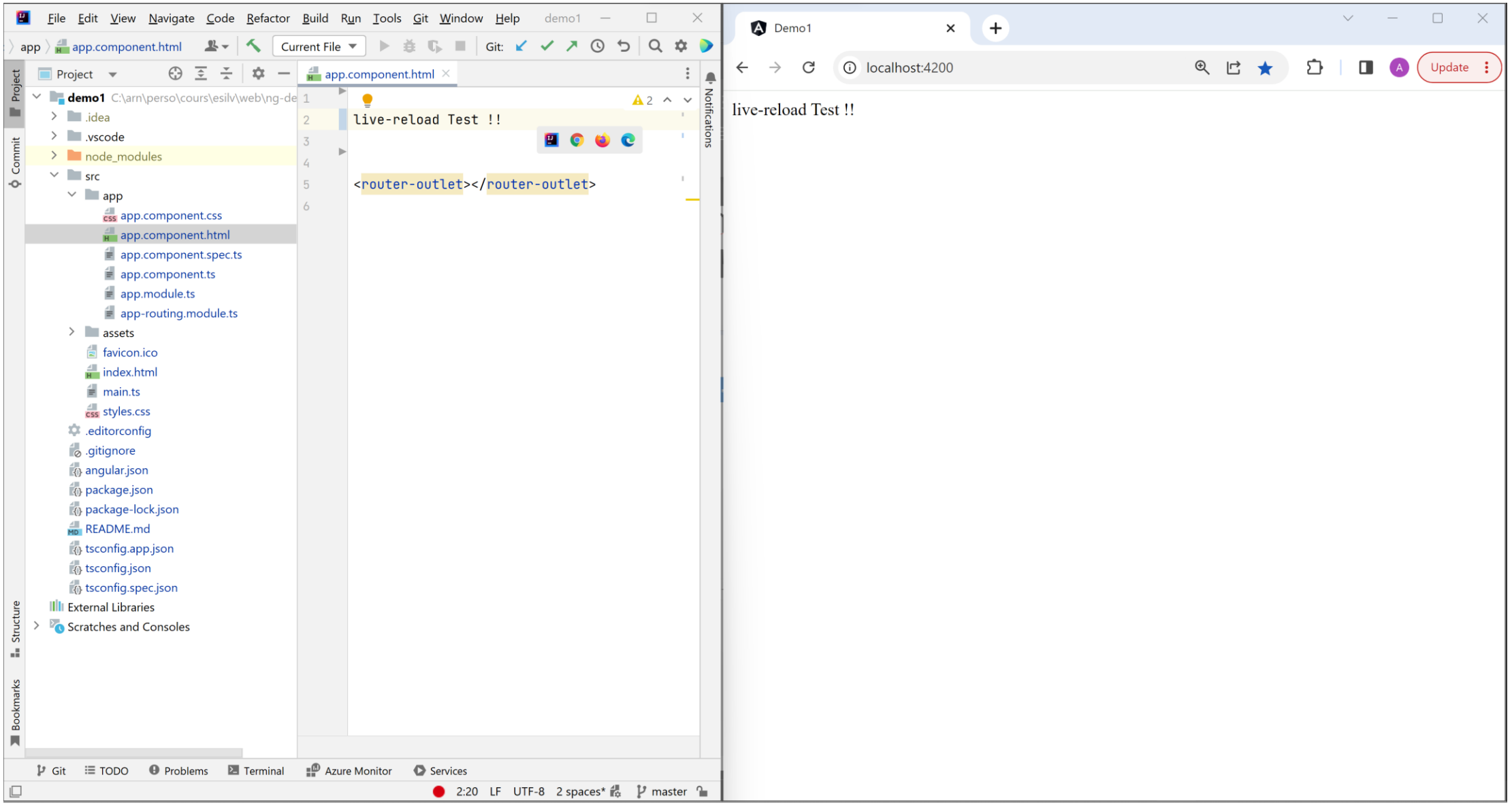
** Angular Live Development Server is listening on localhost:4200, open your browser on http://localhost:4200/ **


√ Compiled successfully.
```

Demo
IntelliJ (left) + Chrome (right) + ng serve

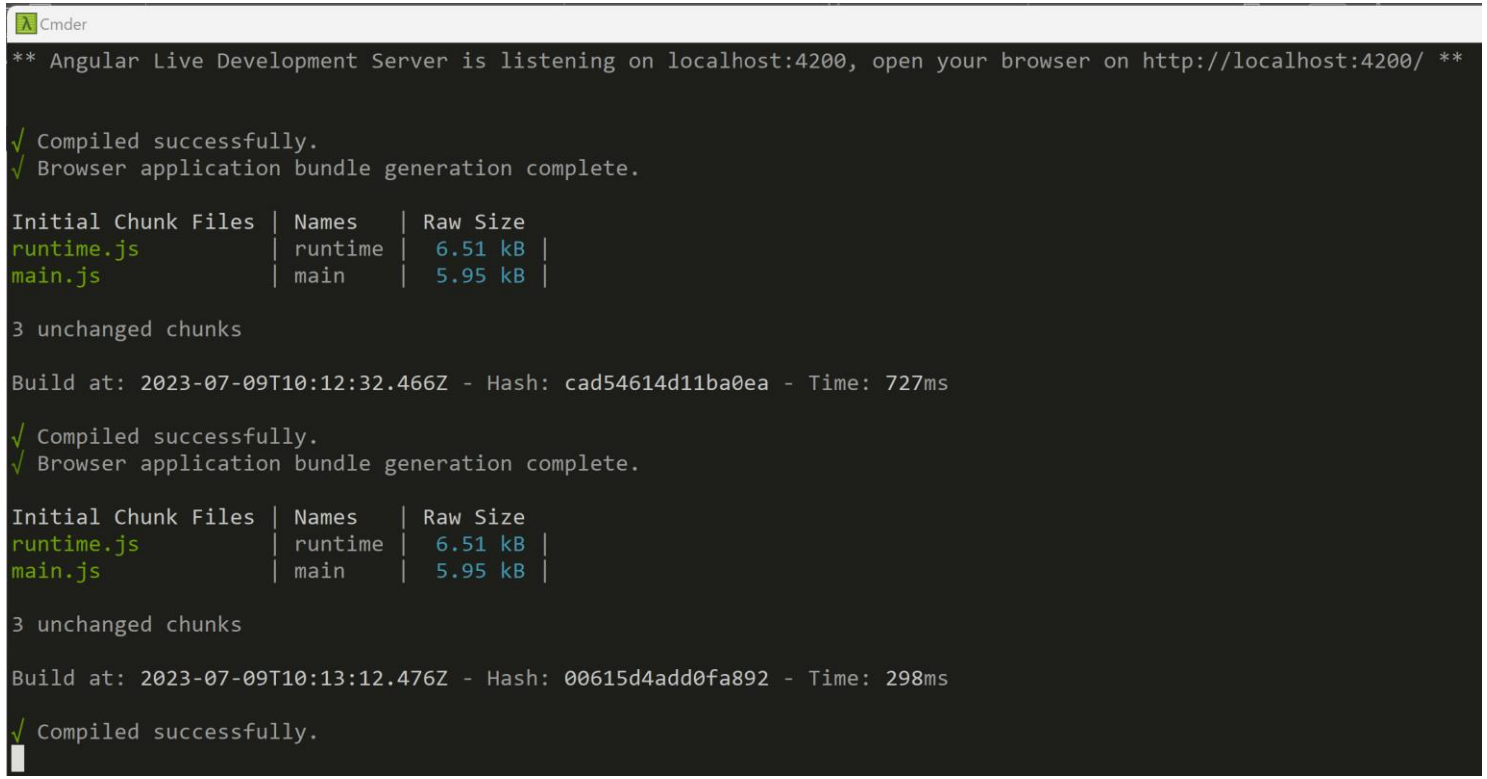


Demo … Live reload

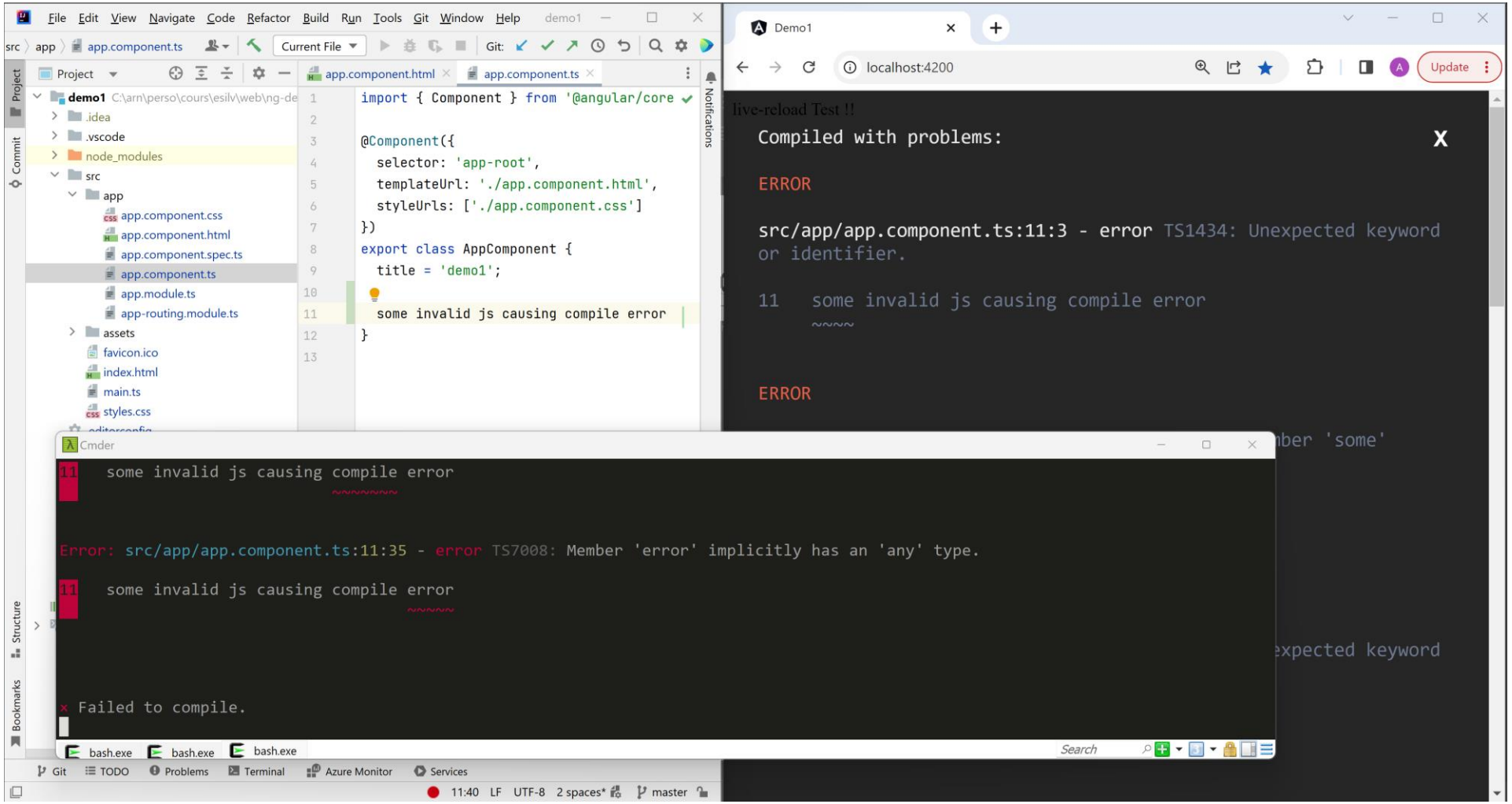


Live Reload … --watch
(expect 5 seconds delay)

watch File changes => compile => refresh page

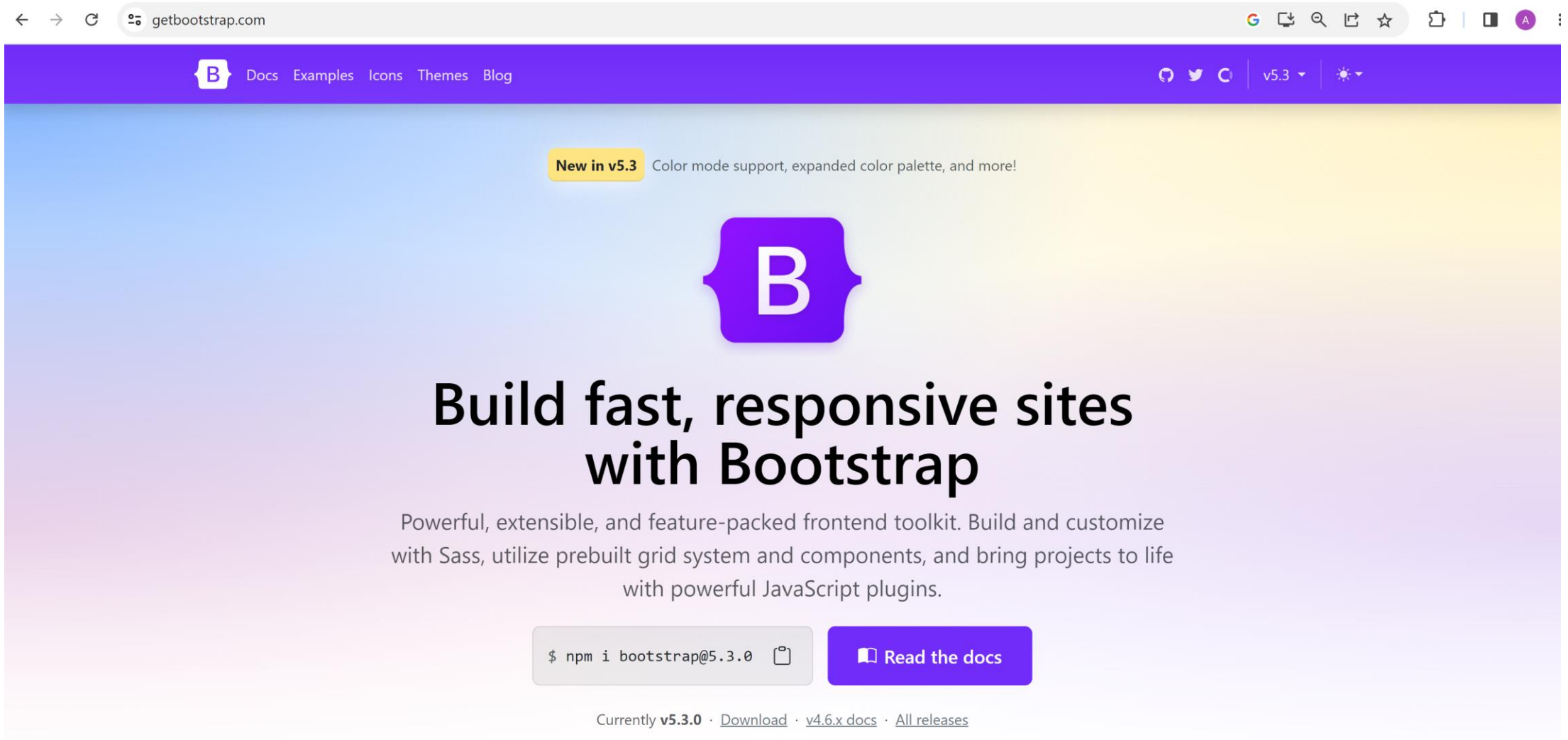


Demo live-reload … when compile error

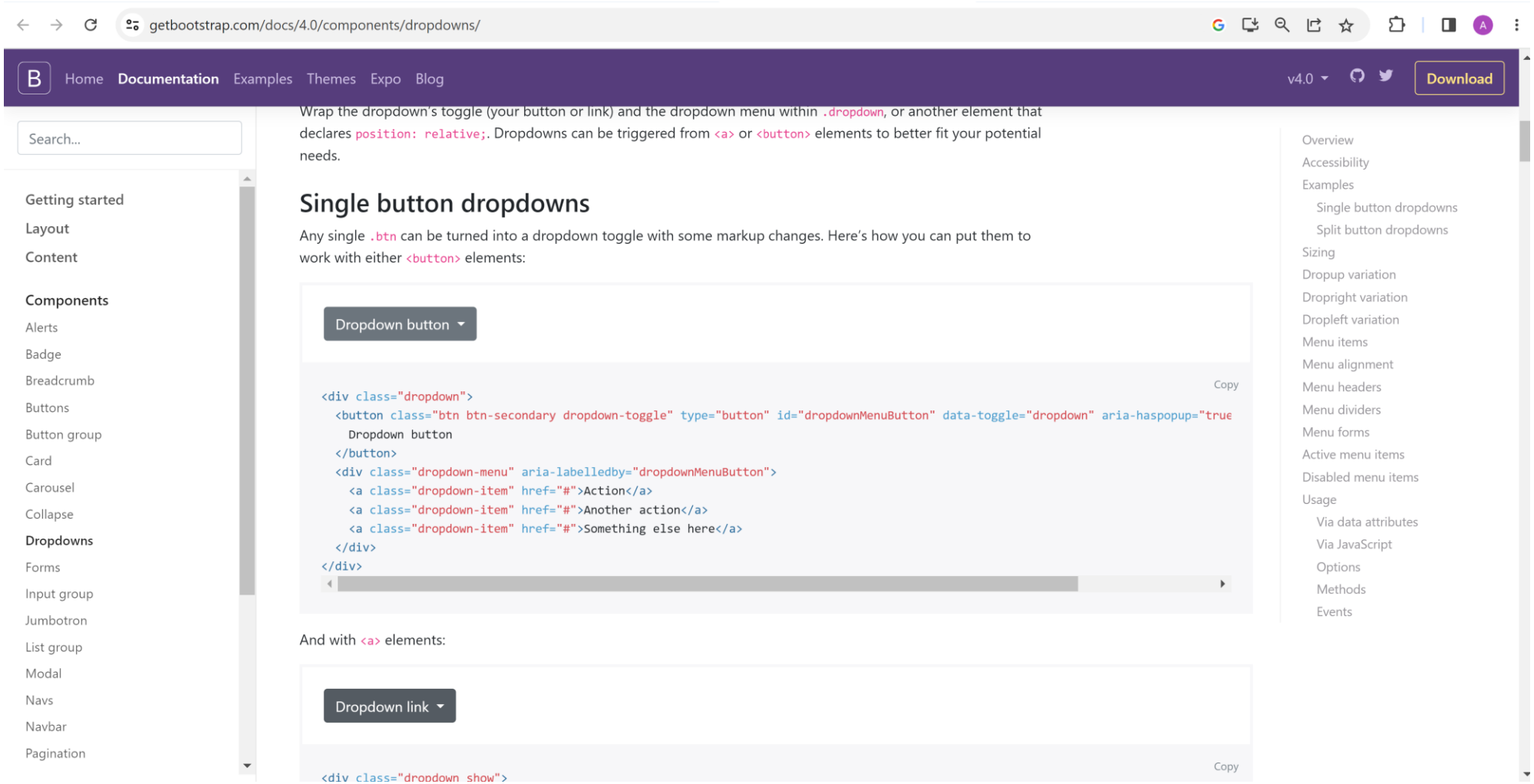


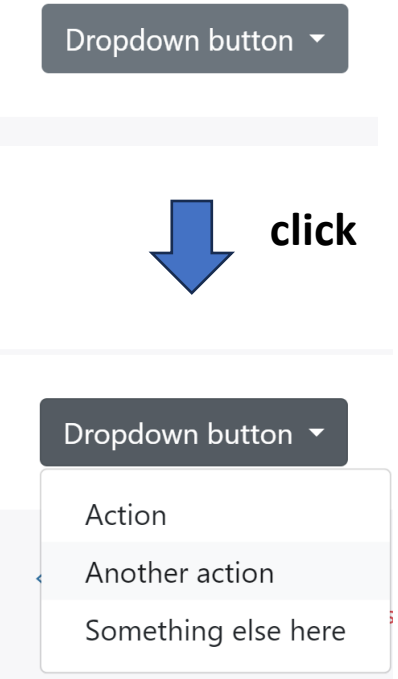
Adding Bootstrap library

https://getbootstrap.com/

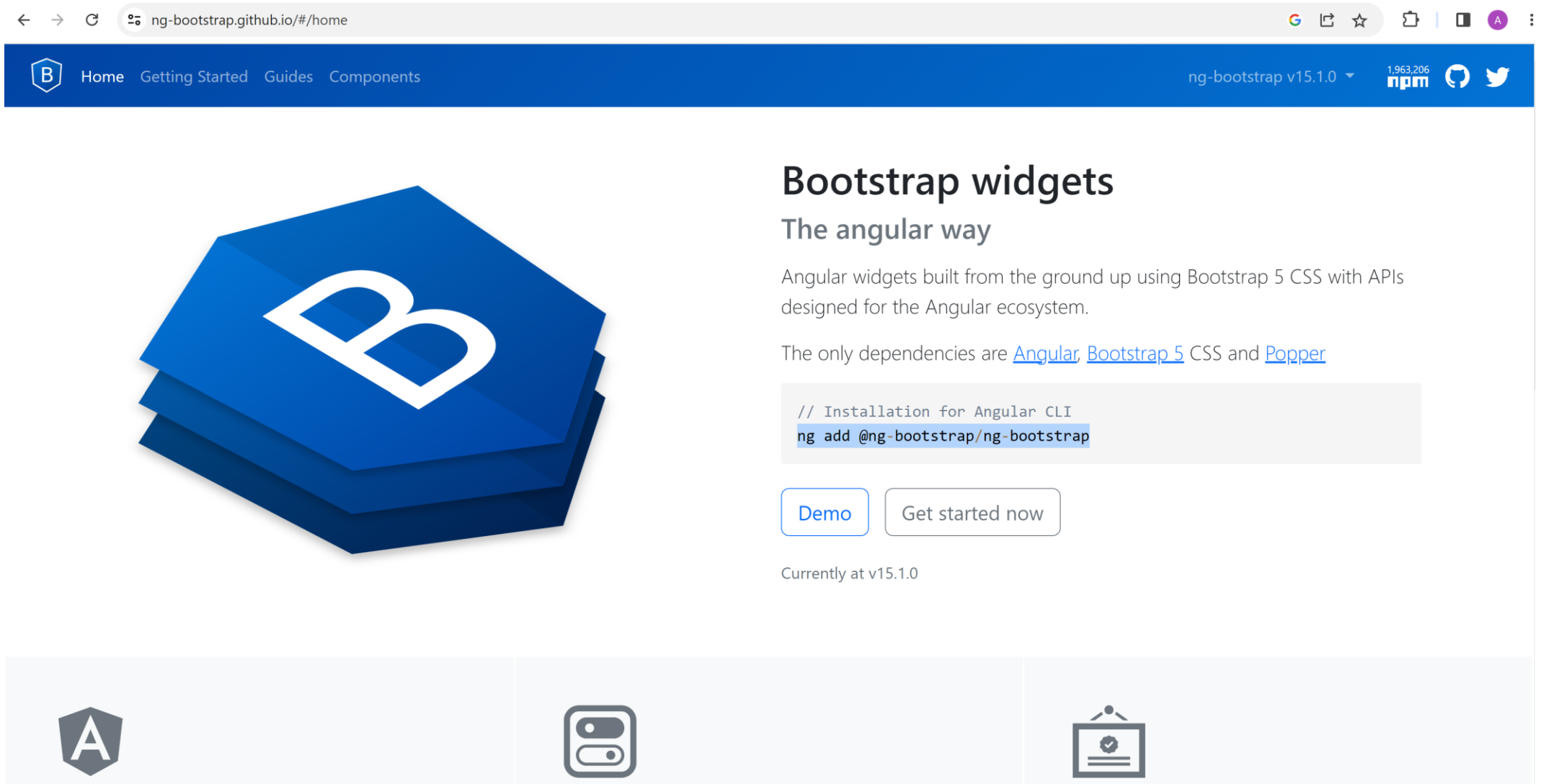


Bootstrap … example: button dropdown





Adding via ng-bootstrap
… NPM module for Bootstrap in Angular



ng add @ng-bootstrap/ng-bootstrap

```
$ ng add @ng-bootstrap/ng-bootstrap
i Using package manager: npm
√ Found compatible package version: @ng-bootstrap/ng-bootstrap@14.2.0.
√ Package information loaded.

The package @ng-bootstrap/ng-bootstrap@14.2.0 will be installed and executed.
Would you like to proceed? Yes
√ Packages successfully installed.
UPDATE package.json (1176 bytes)
√ Packages installed successfully.
Option "name" is deprecated: Use the "project" option instead.
UPDATE src/app/app.module.ts (464 bytes)
UPDATE angular.json (2762 bytes)
UPDATE src/main.ts (259 bytes)
UPDATE tsconfig.app.json (294 bytes)
UPDATE tsconfig.spec.json (300 bytes)
```

git status ; git diff

```
$ git st
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   angular.json
        modified:   package-lock.json
        modified:   package.json
        modified:   src/app/app.module.ts
        modified:   src/main.ts
        modified:   tsconfig.app.json
        modified:   tsconfig.spec.json
```

```
$ git diff
diff --git a/angular.json b/angular.json
index 02ba507..4085657 100755
--- a/angular.json
+++ b/angular.json
@@ -25,6 +25,7 @@
                "src/assets"
              ],
              "styles": [
+               "node_modules/bootstrap/dist/css/bootstrap.min.css",
                "src/styles.css"
              ],
              "scripts": []
diff --git a/package-lock.json b/package-lock.json
index 5e1cf92..f4f0d61 100755
--- a/package-lock.json
+++ b/package-lock.json
@@ -16,6 +16,9 @@
        "@angular/platform-browser": "^15.0.0",
        "@angular/platform-browser-dynamic": "^15.0.0",
        "@angular/router": "^15.0.0",
+       "@ng-bootstrap/ng-bootstrap": "^14.2.0",
+       "@popperjs/core": "^2.11.6",
+       "bootstrap": "^5.2.3",
        "rxjs": "~7.5.0",
        "tslib": "^2.3.0",
        "zone.js": "~0.12.0"
@@ -24,6 +27,7 @@
        "@angular-devkit/build-angular": "^15.0.0",
        "@angular/cli": "~15.0.0",
        "@angular/compiler-cli": "^15.0.0",
+       "@angular/localize": "^15.0.0",
        "@types/jasmine": "~4.3.0",
        "jasmine-core": "~4.5.0",
        "karma": "~6.4.0",
@@ -38,7 +42,6 @@
      "version": "2.2.0",
```

REDO npm install
+ stop & restart ng server

```
$ npm install

removed 1 package, and audited 969 packages in 12s

109 packages are looking for funding
  run `npm fund` for details

22 moderate severity vulnerabilities

To address issues that do not require attention, run:
  npm audit fix

To address all issues (including breaking changes), run:
  npm audit fix --force

Run `npm audit` for details.
```

Bootstrap Demos

Bootstrap demo 1:
Container, 12 x columns system

```
<div class="container">
  <div class="row">
    <div class="col-md-1">col</div>
    <div class="col-md-1">col</div>
    <div class="col-md-1">col</div>
    <div class="col-md-3">col md-3</div>
    <div class="col-md-6">col md-6</div>
  </div>
</div>
```

```
  </div>
  <div class="row">
    <div class="col-md-2 app-border1">col md-2</div>
    <div class="col-md-2 app-border1">col md-2</div>
    <div class="col-md-2 app-border1">col md-2</div>
    <div class="col-md-2 app-border1">col md-2</div>
    <div class="col-md-2 app-border1">col md-2</div>
    <div class="col-md-2 app-border1">col md-2</div>
  </div>
  <div class="row">
    <div class="col-md-3 app-border1">col md-3</div>
    <div class="col-md-3 app-border1">col md-3</div>
    <div class="col-md-3 app-border1">col md-3</div>
    <div class="col-md-3 app-border1">col md-3</div>
  </div>
  <div class="row">
    <div class="col-md-4 app-border1">col md-4</div>
    <div class="col-md-4 app-border1">col md-4</div>
    <div class="col-md-4 app-border1">col md-4</div>
  </div>
  <div class="row">
    <div class="col-md-6 app-border1">col md-6</div>
    <div class="col-md-6 app-border1">col md-6</div>
  </div>
</div>
```

Bootstrap demo 1:
Container, 12 x columns system

Test Bootstrap - 12 columns system

Test Bootstrap class=container, class=row, class=col-*

col col col col md-3 col md-6

... More Test Bootstrap class=container, class=row, class=col-*

col md-1	col md-1	col md-1	col md-1	col md-1	col md-1	col md-1	col md-1	col md-1	col md-1	col md-1	col md-1
col md-2		col md-2		col md-2		col md-2		col md-2		col md-2	
col md-3			col md-3			col md-3			col md-3		
col md-4				col md-4				col md-4			
col md-6						col md-6					

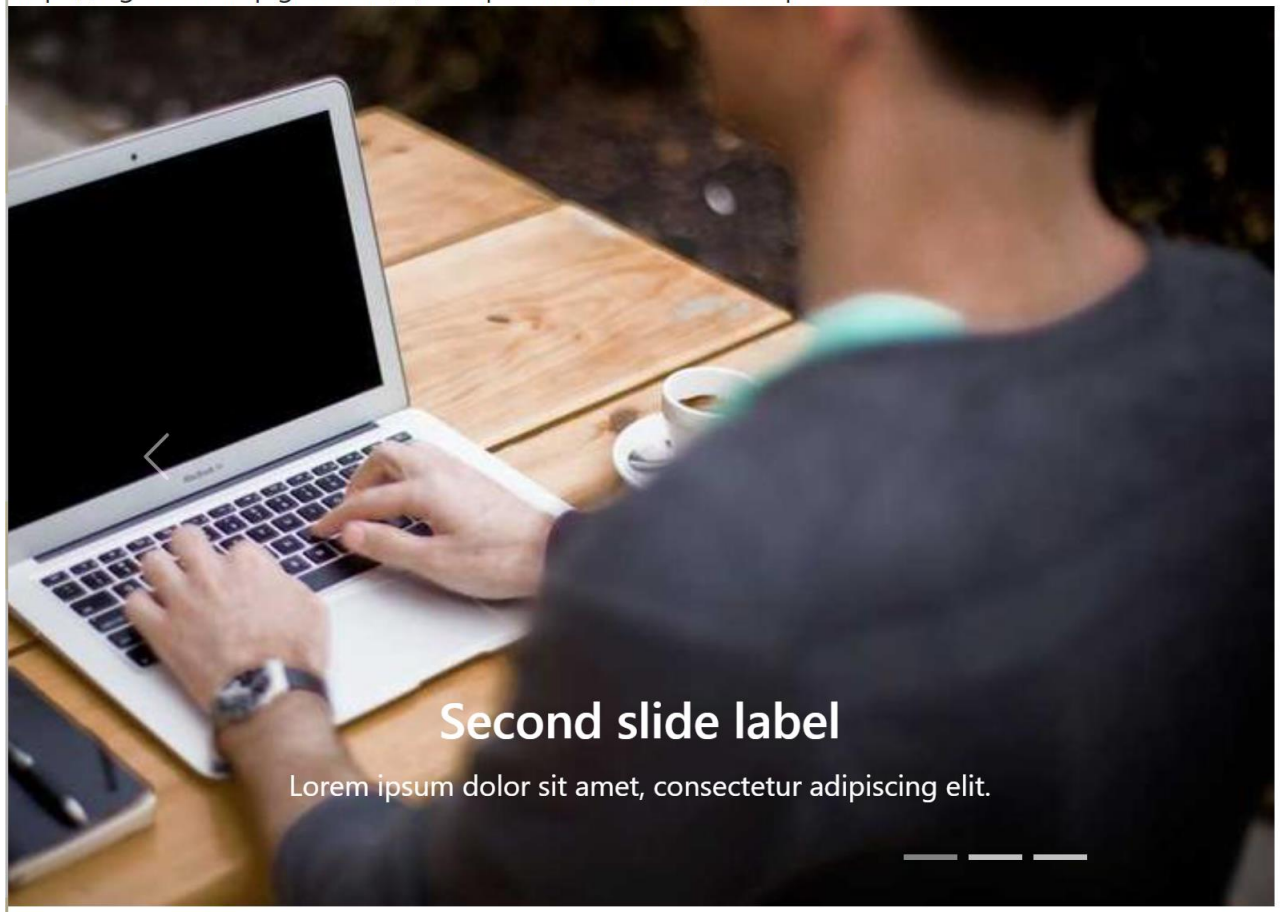
Demo Bootstrap Carousel

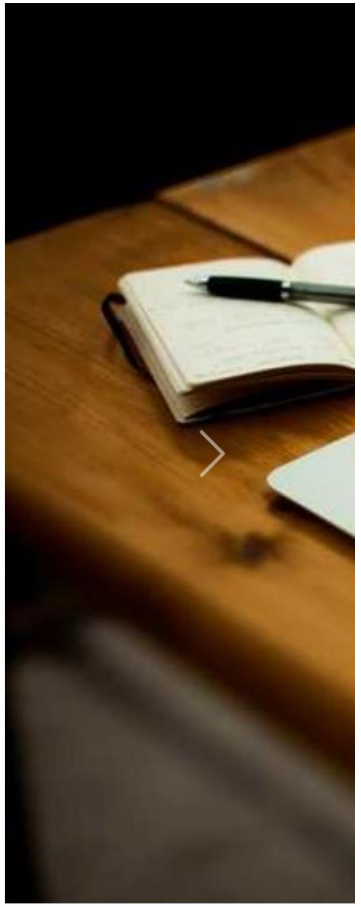
```
<ngb-carousel>
  <ng-template ngbSlide>
    <div class="picsum-img-wrapper">
      <img src="https://picsum.photos/id/0/900/500" alt="Random first slide"/>
    </div>
    <div class="carousel-caption">
      <h3>First slide label</h3>
      <p>Nulla vitae elit libero, a pharetra augue mollis interdum.</p>
    </div>
  </ng-template>
  <ng-template ngbSlide>
    <div class="picsum-img-wrapper">
      <img src="https://picsum.photos/id/1/900/500" alt="Random second slide"/>
    </div>
    <div class="carousel-caption">
      <h3>Second slide label</h3>
      <p>Lorem ipsum dolor sit amet, consectetur adipiscing elit.</p>
    </div>
  </ng-template>
```

Bootstrap demo: Carousel

demo-bootstrap-carousel works!

https://ng-bootstrap.github.io/#/components/carousel/examples





Bootstrap Button

```
<button type="button" class="btn btn-primary">Primary</button>  
<button type="button" class="btn btn-secondary">Secondary</button>  
<button type="button" class="btn btn-success">Success</button>  
<button type="button" class="btn btn-warning">Warning</button>  
<button type="button" class="btn btn-danger">Danger</button>  
```

```
<button type="button" class="btn btn-outline-primary">Primary</button>  
<button type="button" class="btn btn-outline-secondary">Secondary</button>  
<button type="button" class="btn btn-outline-success">Success</button>  
```

```
<button type="button" class="btn btn-primary btn-sm">Small button</button>  
<button type="button" class="btn btn-primary">Std size</button>  
<button type="button" class="btn btn-primary btn-lg">Large button</button>  
```

Bootstrap demo: Button styles

see https://getbootstrap.com/docs/4.0/components/buttons/

button colour: btn-primary, btn-secondary, btn-warning, ..

Primary Secondary Success Warning Danger

button btn-outlined-*

Primary Secondary Success Warning Danger

button size: btn-sm, bt-lg

Small button Std size Large button

Bootstrap demo:
Button dropdown

```
<A href="https://getbootstrap.com/docs/5.0/components/dropdowns/">https://getbootstrap.com/docs

Dropdown button... native class / javascript does not work?!..

<div class="dropdown">
  <button class="btn btn-secondary dropdown-toggle" type="button"
          id="dropdownMenuButton1" data-bs-toggle="dropdown" aria-expanded="false">
    Dropdown button
  </button>
  <ul class="dropdown-menu" aria-labelledby="dropdownMenuButton1">
    <li><a class="dropdown-item" href="#">Action</a></li>
    <li><a class="dropdown-item" href="#">Another action</a></li>
    <li><a class="dropdown-item" href="#">Something else here</a></li>
  </ul>
</div>
```

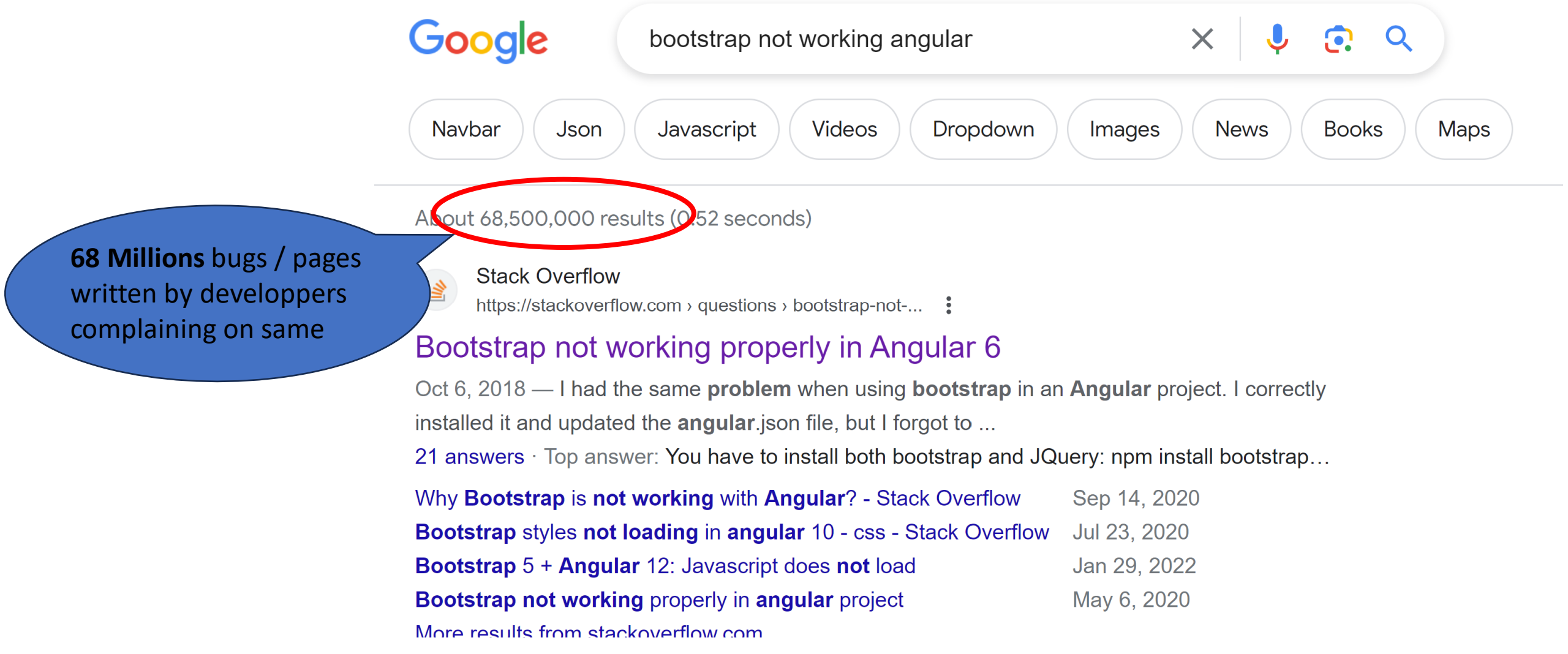
Bootstrap demo:
Button dropdown

https://getbootstrap.com/docs/5.0/components/dropdowns/ Dropdown button... native class / javascript does not work?!..

Dropdown button

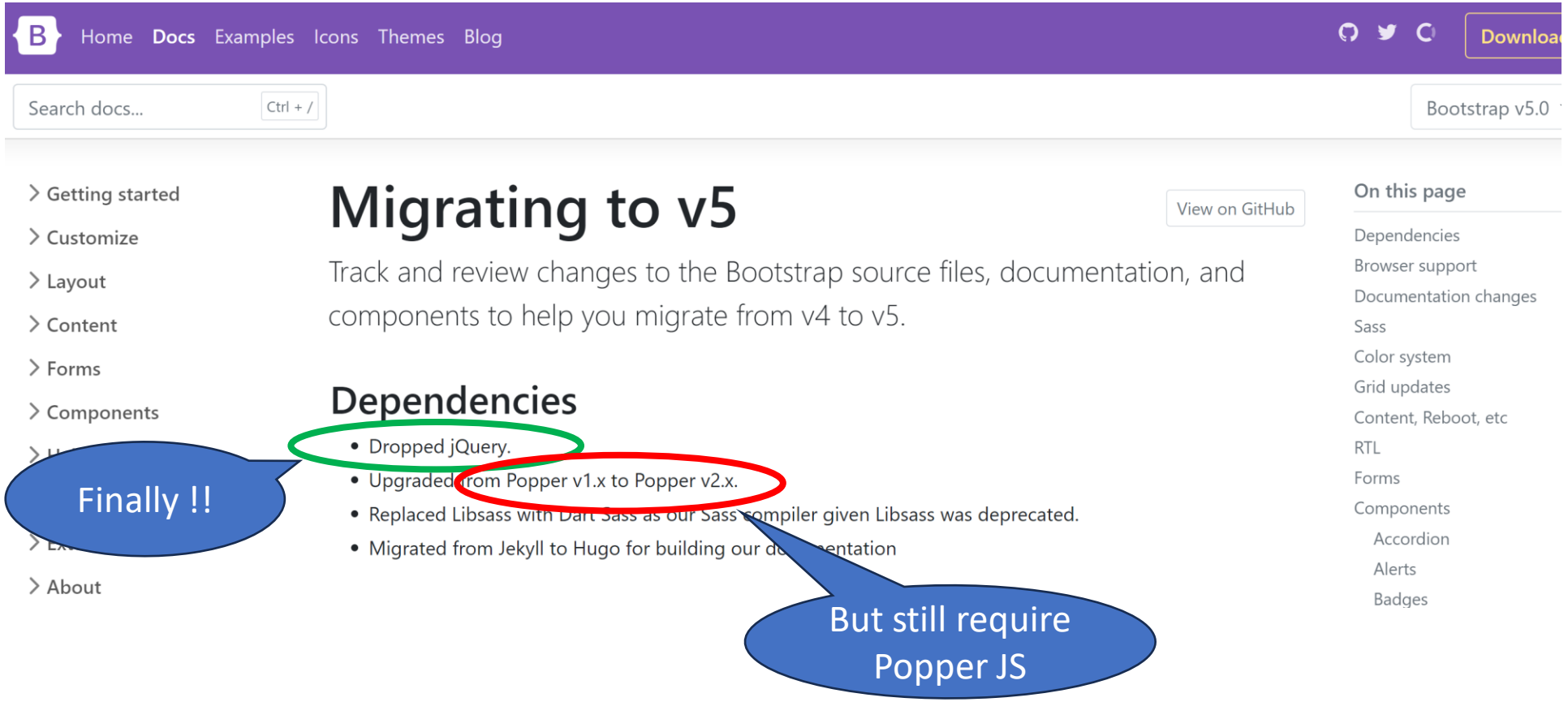
dropdown
does not work?!

!!! Demo may not work ??
You are not alone

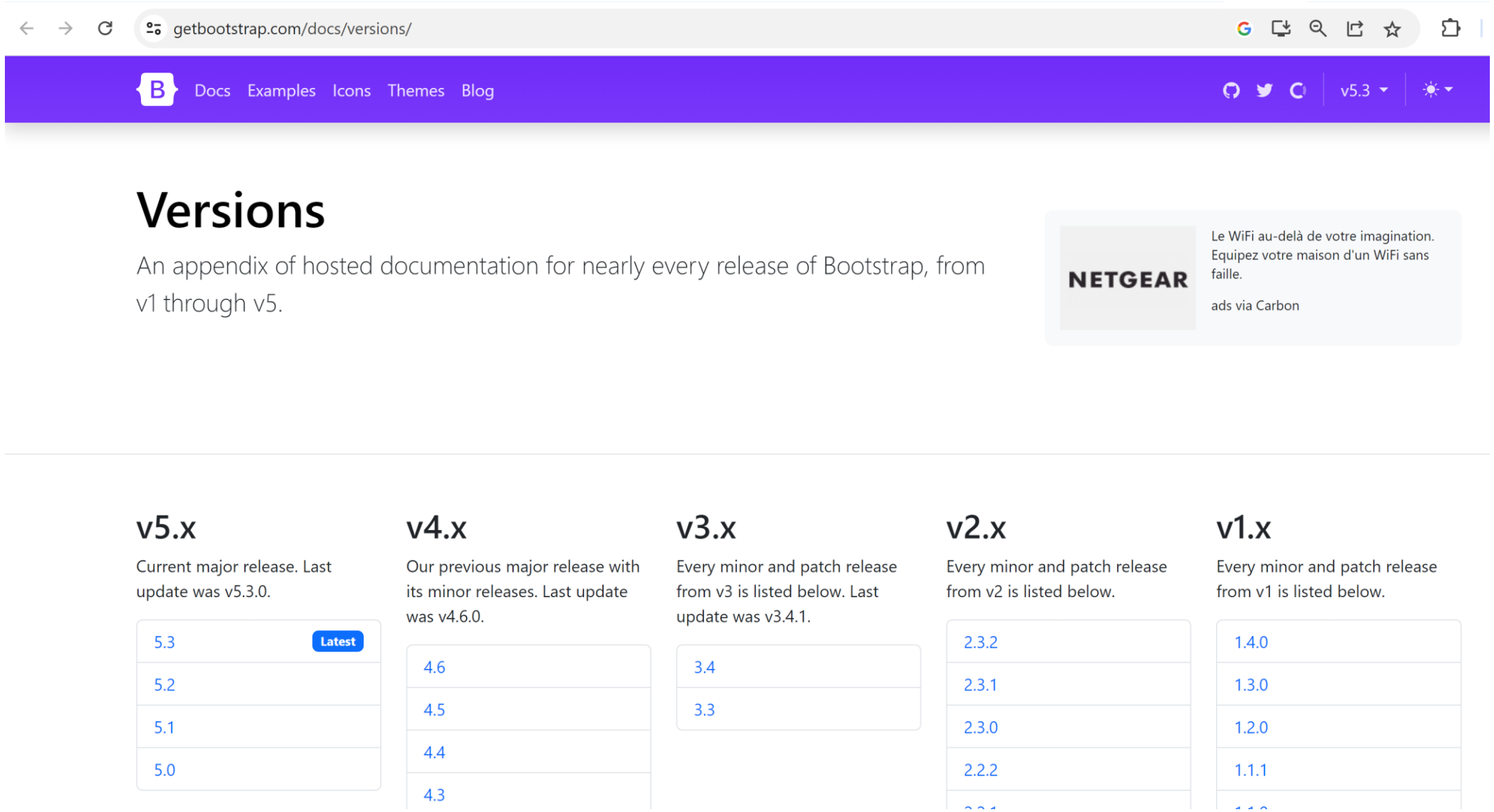


68 Millions bugs / pages written by developpers complaining on same

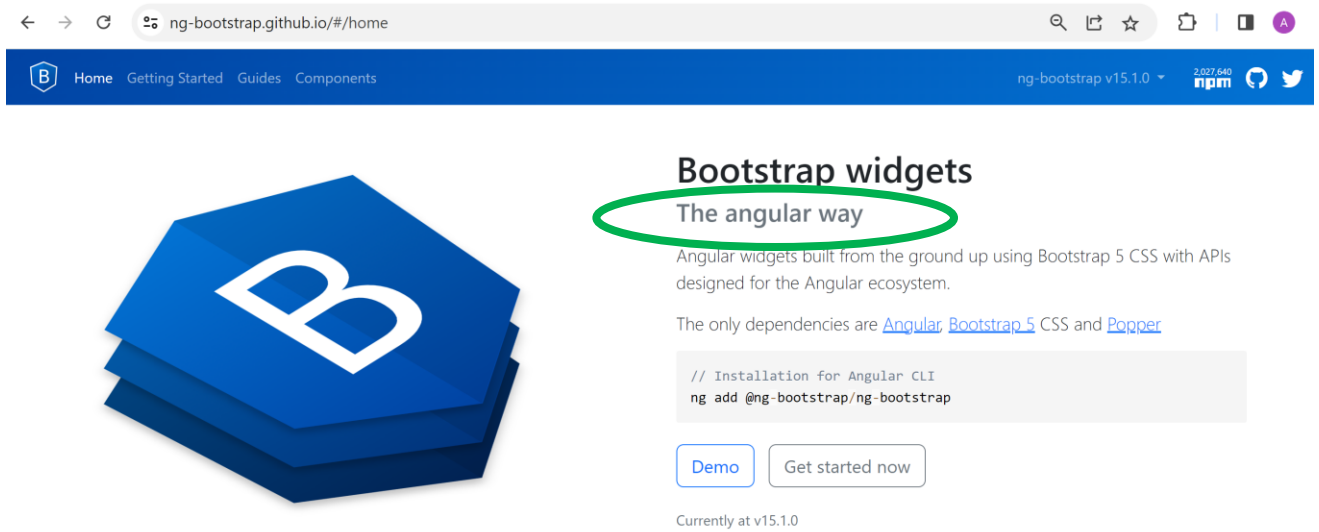
Root cause of all problems: CSS depends on JS



Bootstrap major versions



Testing basic bootstrap with dynamic javascript features => using via « ng-bootsrap »



Some « Vanilla » Bootstrap
Classes + « data-toggle »=.. attributes

Replace by ..

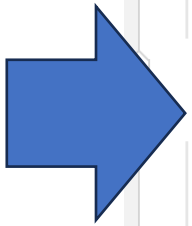
simpler ng-bootstrap equivalent

ngbDropDown
ngbToggle
ngb..

Ngb = initials for
a**ng**ular **B**ootstrap

Bootstrap demo: Button dropdown

```html
<div class="dropdown">
  <button class="btn btn-secondary dropdown-toggle" type="button"
          id="dropdownMenuButton1" data-bs-toggle="dropdown" aria-expa
    Dropdown button
  </button>
  <ul class="dropdown-menu" aria-labelledby="dropdownMenuButton1">
    <li><a class="dropdown-item" href="#">Action</a></li>
    <li><a class="dropdown-item" href="#">Another action</a></li>
    <li><a class="dropdown-item" href="#">Something else here</a></li>
  </ul>
</div>
```



```html
<div ngbDropdown class="d-inline-block">
  <button type="button" class="btn btn-secondary"
          id="dropdownMenuButton2" ngbDropdownToggle>
    Toggle dropdown
  </button>
  <ul ngbDropdownMenu aria-labelledby="dropdownMenuButton2">
    <li><a ngbDropdownItem href="#">Action</a></li>
    <li><a ngbDropdownItem href="#">Another action</a></li>
    <li><a ngbDropdownItem href="#">Something else here</a></li>
  </ul>
</div>
```

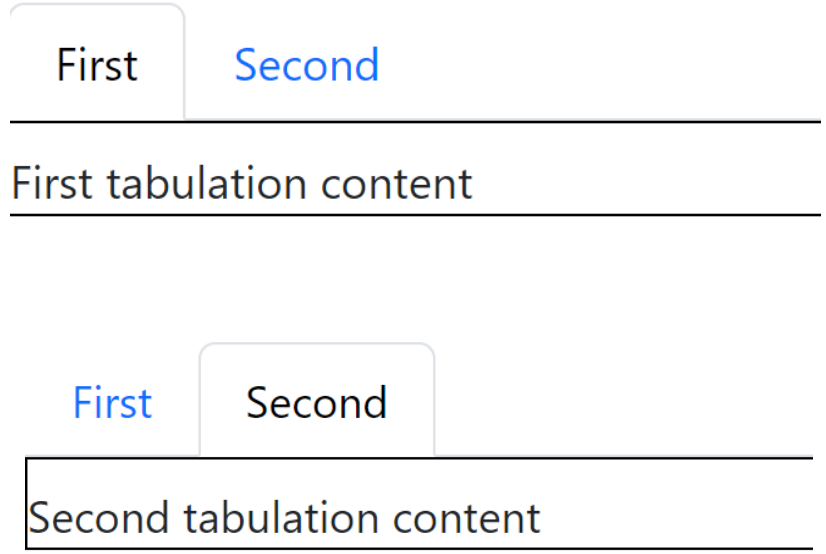
Bootstrap demo:
Button ngbDropdown

data-bs-toggle="dropdown"	-->	ngbDropdown
class="dropdown-toggle"	-->	ngbDropdownToggle
class="dropdown-menu"	-->	ngbDropdownMenu
class="dropdown-item"	-->	ngbDropdownItem



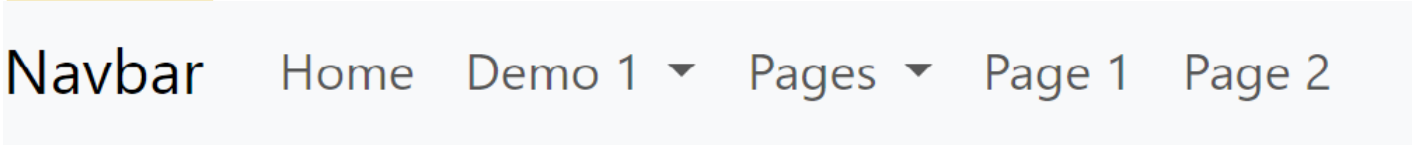
Bootstrap demo: Nav Tabulations

```
<ul ngbNav #nav="ngbNav" class="nav-tabs">
  <li ngbNavItem>
    <button ngbNavLink>First</button>
    <ng-template ngbNavContent>First tabulation content</ng-template>
  </li>
  <li ngbNavItem>
    <button ngbNavLink>Second</button>
    <ng-template ngbNavContent>Second tabulation content</ng-template>
  </li>
</ul>
<div class="app-border1">
  <div [ngbNavOutlet]="nav" class="mt-2"></div>
</div>
```



Bootstrap demo: Navbar Menu

```
<nav class="navbar navbar-expand-lg navbar-light bg-light">
  <a class="navbar-brand" href="#">Navbar</a>
  <ul class="navbar-nav mr-auto">
    <li class="nav-item active">
      <a class="nav-link" href="#">Home</a>
    </li>
    <li ngbDropdown class="nav-item">
      <a ngbDropdownToggle id="navbarMenuDemo1" class="nav-link" role="button">
        Demo 1
      </a>
      <div ngbDropdownMenu aria-labelledby="navbarMenuDemo1" class="dropdown-menu" >
        <a class="dropdown-item" routerLink="/demo1/demo-bootstrap-column">Bootstrap 12 columns</a>
        <a class="dropdown-item" routerLink="/demo1/demo-bootstrap-carousel">Bootstrap Carousel</a>
        <a class="dropdown-item" routerLink="/demo1/demo-bootstrap-button">Bootstrap Button</a>
        <a class="dropdown-item" routerLink="/demo1/demo-bootstrap-button-dropdown">Bootstrap Butto
        <a class="dropdown-item" routerLink="/demo1/demo-bootstrap-navtab">Bootstrap Nav Tabulation
      </div>
    </li>
    <li ngbDropdown class="nav-item">
```



Summary

Install node.js (npm)
$ npm Install –g @angular/cli
$ ng new
=> file: package.json, angular.json
app.component.{html | ts}

$ ng serve
$ ng add @ng-bootstrap/ng-boostrap
$ npm install

<div class=« container »>
<div class=« row »>
<div class=« col-mb-2 »>

<button type=« button » class=« btn btn-primary »>

<nav>

Take Away

$ ng new + ng serve + ng generate + …

=> GET easily a complete Angular environment
… pre-packaging angular + npm + typescript + …

Add (ng-)bootstrap
=> GET basic CSS + elements
… 12-columns layout system
+ button, menu nav bar, tab, etc.

Questions

Next Step

*ngFor
*ngIf
{{interpolation}}
@Component
[] .. @Input (=One way binding)
() .. @Output
[()] .. 2-Ways binding
Router

Document:
https://github.com/Arnaud-Nauwynck/presentations/web/angular-demos-2-ngFor-ngIf-component.pdf